

CV – Alejandro Calleja Luna

Personal Data

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Languages

Spanish - Native
English - Advanced
French - Intermediate

Profile

Senior Engineering Manager combining 9 years of engineering leadership with deep technical foundation from 17 years in software development, progressing from individual contributor to strategic technology leadership.

Hands-on experience informs architectural decisions, performance optimization, technical debt reduction, and AI integration. Expert in distributed systems architecture, design patterns, API design, database optimization, optimizing complex SQL queries, ETL pipeline design, microservices modernization, and legacy code transformation.

Comfortable making software and solutions architecture decisions, reviewing code, and mentoring engineers on best practices. Successfully led agile methodologies adoption (Scrum/Kanban), ISO 27001 certification, and remote work transition.

Seeking Software Development Manager, Technical Lead, or Solutions Architect roles where technical depth, artificial intelligence experience, and proven leadership create competitive advantage.

Job Experience

2025 July – Present

Aduanapp – AI Customs Tech Startup
AI & Data Engineer (Part-time)

- **AI Integration and Development**
 - Evaluated and integrated third-party APIs to enhance product capabilities and data enrichment.
 - Analyzed technical feasibility of new features and strategic initiatives for AI-powered customs digitalization platform.
 - Integrated AI tools to accelerate the software development life cycle.
 - Configured CI/CD pipelines with GitHub Actions: automated linting, testing, and branch protection in production.
 - Designed data analysis pipelines for intelligence and gap identification.

2020 January – 2025 May

ITCE – Innovaciones Tecnológicas para el Comercio Exterior
Technology Director (Senior Engineering Manager)

- **Technology Direction and Strategy**
 - Defined and implemented technology strategy aligned with business objectives, developing innovative software solutions that automated workflows and centralized data management for systems processing ~100K monthly transactions.
 - Oversaw technology budget, optimizing cloud infrastructure and development resources, reducing operational costs by ~10%.
 - Implemented 12-16 annual OKRs and 3 KPIs across development and infrastructure, improving visibility and decision-making.
- **Software Development and Architecture**
 - Expanded and strengthened communication infrastructure, adding 1 new bank and 3 redundancy channels (total: 2 SAT, 6 banks, 3 redundant), improving system redundancy.
 - Optimized high-performance ETL pipelines (~100K monthly records).
 - Modernized 5-6 legacy applications annually through microservices architectures and scalable design patterns.

CV – Alejandro Calleja Luna

- Adopted CI/CD pipelines enabling 20-30 monthly deployments with version control and formal release process.
- Explored and evaluated artificial intelligence and machine learning solutions (GCP Vertex AI) to enhance system capabilities and automation.
- **Team Leadership and Organizational Culture**
 - Led 4-person team (2 Senior Developers, 1 Junior Developer, 1 Senior Infrastructure Engineer), delivering 8-12 annual projects.
 - Championed adoption of agile methodologies and engineering best practices, improving delivery speed and fostering collaborative team environments that enabled successful remote work transition since 2019.
 - Continuously evaluated emerging technologies.
- **Project Management**
 - Oversaw 4-6 simultaneous projects, managing costs, risks, and deliveries.
 - Designed quality control and incident management processes for critical systems that scaled from ~60K to ~120K monthly transactions.
 - Led migration of critical infrastructure (3 servers, 3 databases) to Google Cloud Platform, improving availability from 97% to 99% in 6 months.
- **Information Security**
 - Implemented and continuously evolved Information Security Management System (ISMS): achieved ISO 27001:2013 certification and successfully transitioned to ISO 27001:2022 in 2024, maintaining compliance.
 - Coordinated ISO 27001 audits (2 internal, 1 external annually) since 2020, implementing actions for all non-conformities and observations.
 - Managed security risks, identifying, assessing, and mitigating potential threats, implementing controls that reduced critical vulnerabilities.
- **Suppliers Management**
 - Built strategic relationships with SAT, Banks in Mexico, clients, and suppliers for technical alignment.
 - Revised 20+ supplier contracts (3 critical) and client agreements, validating compliance with technical specifications and monitoring SLAs.

2016 June – 2019 December

ITCE – Innovaciones Tecnológicas para el Comercio Exterior
Development Subdirector (Engineering Manager)

- **Business Intelligence**
 - Created ~40 reports and dashboards for systems processing ~60K monthly transactions, for Finance, Operations, Executive Management, and clients.
 - Analyzed data for strategic company decision-making.
- **Software Development and Architecture**
 - Made key contributions to Electronic Pre-validation software in Mexico (10-15 core processes), modernizing legacy systems and later expanding to 30+ processes.
 - Orchestrated complete development lifecycle: requirements gathering, analysis, design, development, testing, deployment, and continuous improvement.
 - Conducted ~10 weekly code reviews and established coding standards, ensuring clean, efficient, and well-documented code adhering to best practices.
- **Project Management**
 - Led 3-developer team (2 Senior, 1 Junior) managing 2-3 simultaneous projects, with average delivery of 4 annual projects.
 - Introduced agile methodologies (Scrum/Kanban) that improved delivery times and enabled team participation across multiple projects.
- **Information Security**
 - Contributed to ~60 technical security controls (policies, processes, procedures) for ISO 27001 preparation, participating in 15+ audits.
- **Suppliers Management**
 - Established, tested, and monitored 7 critical communication channels (2 with SAT, 5 with banks via VPN/MPLS), to enable secure information exchange.

CV – Alejandro Calleja Luna

2008 June – 2016 May

ITCE – Innovaciones Tecnológicas para el Comercio Exterior
Software Engineer

- **Software Development and Architecture**
 - Redesigned customs validation system architecture, transforming monolithic process into distributed architecture, reducing processing times from 40 minutes to 5 minutes during peak hours (~3K-4K daily files).
 - Developed Java backend applications for business rule validation according to Mexican customs regulations.
 - Significantly reduced support incidents through system response time improvements.
- **Data Engineering**
 - Designed and implemented from scratch a centralized Data Warehouse solution consolidating information from 3 distributed databases across multiple servers, establishing global standards for data management and integration.
 - Built automated ETL processes with daily updates (D-1) for data loading and transformation into Data Warehouse.
 - Architected multi-server synchronization using custom scripts (rsync + dumps) and Java processes for data loading.
 - Normalized 30+ database tables, defined primary/foreign keys, and implemented strategic indexes.
- **Optimization and automation**
 - Streamlined SQL queries in MySQL, reducing execution times from minutes to seconds through indexes, normalization, and materialized views.
 - Integrated connection pooling and workflow automation in critical processes, eliminating connection exhaustion issues and establishing best practices for resource management and code quality.
 - Served as technical reference for the team on database performance, query optimization, and ETL architecture topics.
- **Monitoring and support**
 - Monitored and resolved incidents in production systems through log analysis and database troubleshooting.
 - Continuously improved system reliability, reducing bottlenecks and single points of failure.

Education

Universidad La Salle México

[2004-2008] Cybernetics and Computer Systems Engineering

[2011-2013] Master's in Information Technology for Business Management

Knowledge

Languages: Java, C/C++, C#, Java EE, JavaScript, shell, Perl, TypeScript, Python

Web: HTML, HTML5, CSS, JSP, XML, JSON, JQuery, Angular, React, Next.js, Tailwind CSS

Frameworks: Spring, Bootstrap, Hibernate, Maven, Apache Tomcat, JBoss, Node.js, Express, FastAPI, Vite

Databases: SQL, MySQL, Oracle, DB2, Power BI, ETL, Business Intelligence, data warehouse, PostgreSQL, ClickHouse, DuckDB, Supabase, star & snowflake schemas

Cloud: GPC (SQL, BigQuery, Compute Engine, Storage, VPN, Buckets, Vertex AI, Dataflow, Composer, Pub/Sub, Functions), AWS (instances, lambda, s3, redshift, rds, vpc), Railway

Operating systems: Windows, macOS, CentOS, Red Hat

Version control: Git, GitHub, GitKraken

Tools: Eclipse, NetBeans, IntelliJ, Visual Studio, SoapUI, Postman, GitHub Actions

Architecture: MVC, DAO, Facade, Singleton, Microservices, OOP, REST API.

Methodologies & management: Scrum, Kanban, ITIL v3, BPM, TOGAF

Standards & certifications: ISO 20000, ISO 27001:2013/2022, ISO 27002, ISO 27301

Skills: Lifelong learning, leadership, effective communication, teamwork, decision-making
